

Unit 5, Lesson 7: Corresponding Parts of Congruent Triangles Are Congruent



Benchmarks

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I understand how to use corresponding parts of congruent triangles in a proof.
- ☐ I can prove parts of triangles are congruent using triangle congruence and corresponding parts of congruent triangles.



5.7.1 Warm-Up: Brain Teaser

Brain teasers like this have been traced back to medieval times. Here's one with a Florida twist.

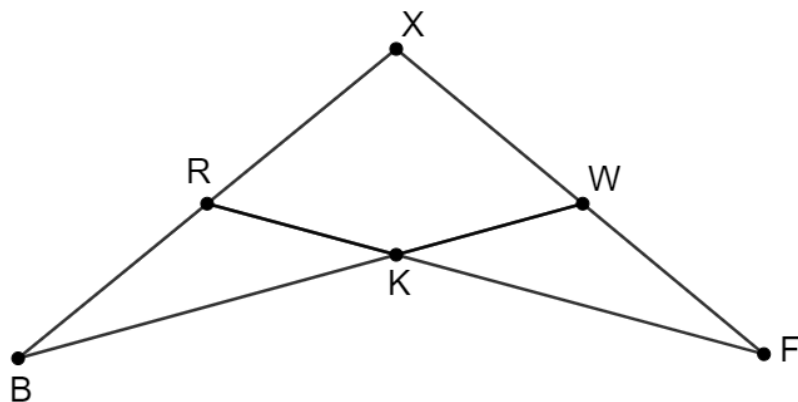
1. A man has to take a panther, a raccoon, and some strawberries across St. John's river. His rowboat has enough room for the man, plus either the panther or the raccoon or the strawberries. If he takes the strawberries with him, the panther will eat the raccoon. If he takes the panther, the raccoon will eat the strawberries. Only when the man is present are the raccoon and the strawberries safe from their enemies. All the same, the man successfully carries panther, raccoon, and strawberries across the river. How?



5.7.2 Collaboration: Overlapping Triangles

Work with your partner to complete the following.

Polygon $XFKB$ is shown.



1. How many triangles are in polygon $XFKB$?
2. Draw $\triangle XBW$ and $\triangle XFR$ separately. Label where point K is on each triangle.

3. Discuss with your partner if triangles $\triangle XBW$ and $\triangle XFR$ have any common sides or angles. If so, name the common part(s).

4. Explain why knowing if $\triangle XBW$ and $\triangle XFR$ have any common sides or angles could be helpful in a proof.

5. In your drawing of $\triangle XBW$ and $\triangle XFR$ as separate triangles, mark the common part(s) as congruent.

6. What property justifies that something is equal to itself?

7. Mark the given information in your diagram.

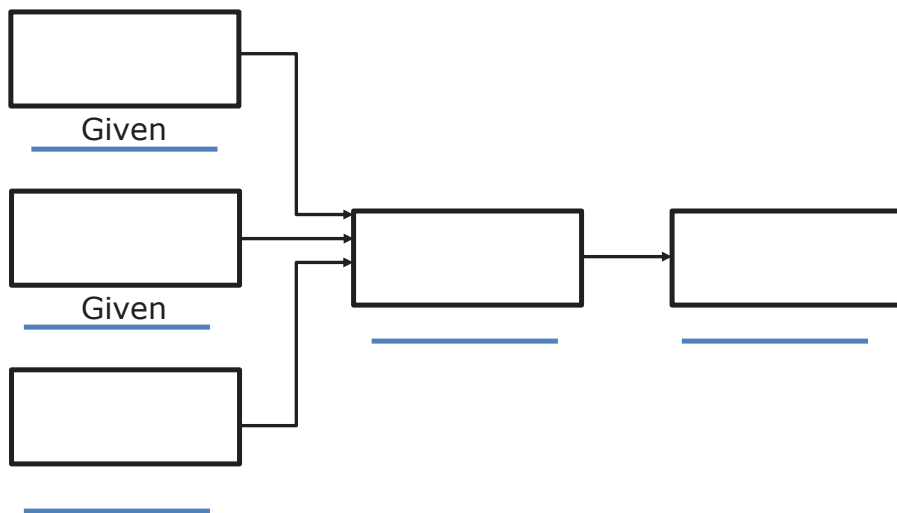
Given: $\overline{RX} \cong \overline{WX}$, $\angle XRK \cong \angle XWK$

8. What can you conclude about $\triangle XBW$ and $\triangle XFR$ from the information you marked on your drawing?

9. Complete the flowchart proof for polygon $XFKB$.

Given: $\overline{RX} \cong \overline{WX}$, $\angle XRK \cong \angle XWK$

Prove: $\angle XBK \cong \angle XFK$



10. Check your proof with another group and if necessary, make adjustments.



5.7.3 Guided Instruction: Proofs Using CPCTC

Once two triangles have been proven to be congruent, then CPCTC states that any other corresponding parts of the triangles are also congruent.

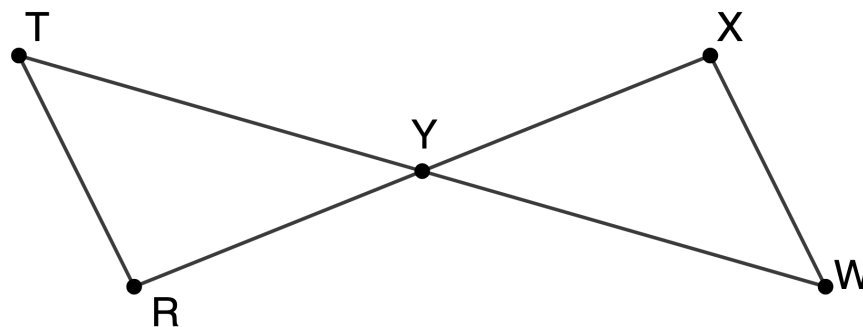


Corresponding Parts of Congruent Triangles are Congruent (CPCTC)

If two triangles are congruent, then all corresponding parts of the triangles are congruent.

In the flowchart proof, proving that $\angle XBK \cong \angle XFK$ first required proving that $\triangle XBW$ and $\triangle XFR$ are congruent.

1. Triangles TYR and XYW are shown.



a. Mark the diagram to show the following given information.

Given: Y is the midpoint of $\overline{TW}$. Y is the midpoint of $\overline{RX}$.

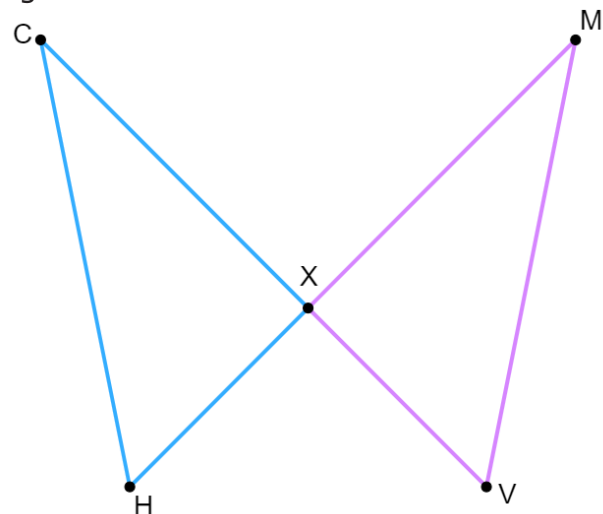
- b. What does “Y is the midpoint of $\overline{WT}$ ” allow you to conclude about the diagram?
- c. What does “Y is the midpoint of $\overline{RX}$ ” allow you to conclude about the diagram?
- d. Discuss with your partner if there is any other information about the two triangles that can be determined from the diagram. Write a summary of your discussion.
- e. Work with your partner to complete the paragraph proof.

Given: Y is the midpoint of $\overline{WT}$. Y is the midpoint of $\overline{RX}$.
 Prove: $\overline{TR} \cong \overline{XW}$

It is given that Y is the midpoint of $\overline{WT}$, so _____ by _____. Also, it is given that Y is the midpoint of $\overline{RX}$, so _____ by _____. Therefore by _____, Δ _____ $\cong$ Δ _____ so by _____, $\overline{TR} \cong \overline{XW}$.



2. Triangles CXH and XMV are shown.



Given: $\overline{CV} \perp \overline{HM}$, $\overline{HX} \cong \overline{XV}$, $\overline{HC} \cong \overline{MV}$
 Prove: $\overline{XC} \cong \overline{XM}$

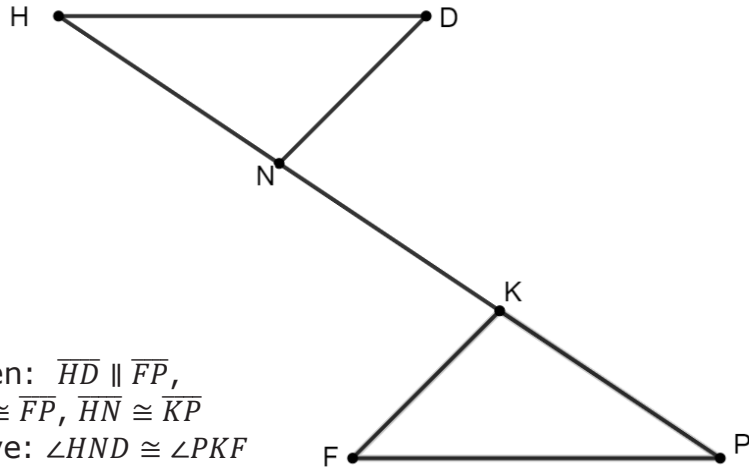
- a. Mark the diagram to show the given information.
- b. Which congruence theorem or postulate do you think should be used?
- c. Complete the two-column proof.

Statement	Reason
1. $\overline{CV} \perp \overline{HM}$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.
7. $\overline{XC} \cong \overline{XM}$	7.



5.7.4 Guided Practice: Proofs Using CPCTC

1. A diagram of triangles NDH and PKF with $\overline{NK}$ is shown.



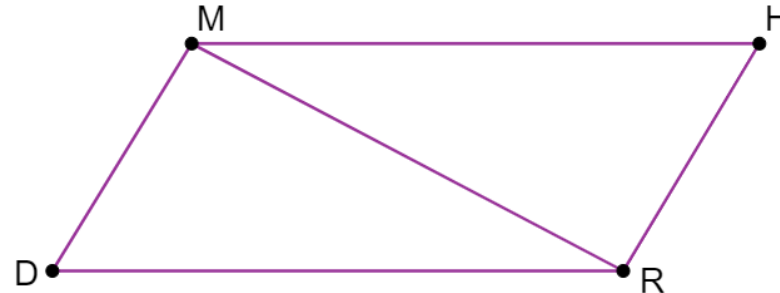
Complete the paragraph proof.

It is given that $\overline{HD} \parallel \overline{FP}$, so by _____,
 $\angle _____ \cong \angle _____$. Since it is given that $\overline{HD} \cong \overline{FP}$ and
 $\overline{HN} \cong \overline{KP}$, $\Delta _____ \cong \Delta _____$ by _____. Therefore,
 $\angle HND \cong \angle PKF$ by _____.



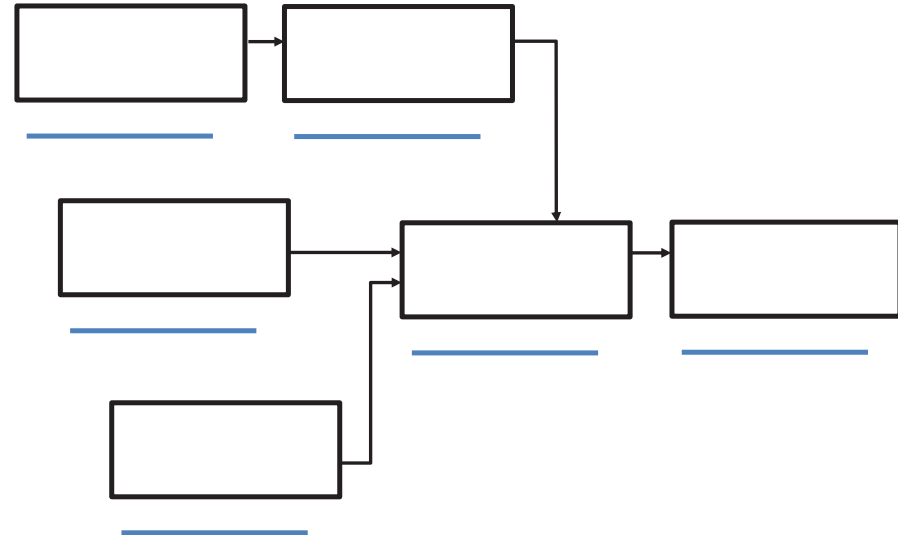
5.7.5 Your Turn!

1. Quadrilateral $MHRD$ is shown.



Given: $\overline{DM} \parallel \overline{RH}$, $\angle RDM \cong \angle RHM$
 Prove: $\overline{DM} \cong \overline{RH}$

Complete the flowchart proof.





5.7.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.

1. I understand how to use corresponding parts of congruent triangles in a proof.

I need help. I can't get started. I'm getting there, but I need more practice. I understand this and I feel confident.



2. I can prove parts of triangles are congruent using triangle congruence and corresponding parts of congruent triangles.

I need help. I can't get started. I'm getting there, but I need more practice. I understand this and I feel confident.



Unit 5, Lesson 8: The Proof Is in the Bag!



Benchmarks

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MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can prove two triangles are congruent using SSS, SAS, ASA, AAS, or HL congruence.
- ☐ I can prove parts of triangles are congruent using triangle congruence and corresponding parts of congruent triangles.